

PAPER_1328 — Galaxy Morphology n=4 Main Types

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "galaxy_morphology"})`

Master Expression

$n_types = D_phys = 4$ (E, S, Irr, dwarf); $subtypes = D_phys \times D_BSFG = 24$

Match: STRUCTURAL

Interpretation

Hubble sequence emerges from $D_phys = 4$ main + $D_BSFG = 6$ subtypes per family.

UQFF Locked Primitives

- $D_phys = 4$, $D_BSFG = 6$, $D_crit = 26$, $SO_5 = 10$, $A_5 = 60$, $N_CH = 9$
- $K_MEX = 25/12$, $\beta_i = 0.6029$, $\Phi_res = 0.84$, $\Lambda = 0.00729735$, $F_TRZ = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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